

Programming Fundamentals – Lab 05

Tasks

Question 1: Gym Membership Discount Evaluation (Nested If-Else)

A fitness center offers membership discounts based on fitness level and age:

- If Fitness Score ≥ 85
 - If Age $< 25 \rightarrow 40\%$ Discount
 - Else $\rightarrow 25\%$ Discount
- Else
 - If Fitness Score ≥ 70 AND Age $< 30 \rightarrow 15\%$ Discount
 - Else $\rightarrow$ No Discount

✦ Task:

Write a C program using nested if-else statements and logical operators to determine the discount for a member.

Code:

```
int main() {  
    int fitnessScore, age;  
  
    printf("Enter fitness score: ");  
    scanf("%d", &fitnessScore);  
  
    printf("Enter age: ");  
    scanf("%d", &age);  
  
    if (fitnessScore >= 85) {  
        if (age < 25) {  
            printf("Discount: 40%%\n");  
        } else {
```

```

        printf("Discount: 25%%\n");
    }
} else {
    if (fitnessScore >= 70 && age < 30) {
        printf("Discount: 15%%\n");
    } else {
        printf("No Discount\n");
    }
}

return 0;
}

```

Question 2: Mobile Data Billing System (Nested If-Else + Logical Operators)

A telecom company charges for data usage as follows:

- If Data $\leq$ 2 GB $\rightarrow$ Rs. 50 per GB
- If Data $>$ 2 GB and $\leq$ 10 GB
 - If Customer Type = Prepaid $\rightarrow$ Rs. 40 per GB
 - If Customer Type = Postpaid $\rightarrow$ Rs. 35 per GB
- If Data $>$ 10 GB $\rightarrow$ Rs. 25 per GB

✦ Task:

Design a C program using nested decision structures to calculate and display the total mobile data bill.

Code:

```

int main() {
    float dataUsed, bill = 0;
    int customerType;

    printf("Enter data used (in GB): ");
}

```

```
scanf("%f", &dataUsed);

if (dataUsed <= 2) {
    bill = dataUsed * 50;
}
else if (dataUsed <= 10) {
    printf("Enter customer type (1 = Prepaid, 2 = Postpaid): ");
    scanf("%d", &customerType);

    if (customerType == 1) {
        bill = dataUsed * 40;
    } else if (customerType == 2) {
        bill = dataUsed * 35;
    } else {
        printf("Invalid customer type\n");
        return 0;
    }
}
else {
    bill = dataUsed * 25;
}

printf("Total Mobile Bill: Rs. %.2f\n", bill);

return 0;
}
```

Question 3: Airport Luggage Fee Calculator (Nested Switch-Case)

An airport charges luggage fees based on:

Main Luggage Type:

1 → Handbag

2 → Checked Bag

3 → Sports Equipment

- If Handbag:
 - 1 → Small (Rs. 100 per kg)
 - 2 → Large (Rs. 200 per kg)
- If Checked Bag:
 - 1 → Domestic (Rs. 300 per kg)
 - 2 → International (Rs. 500 per kg)
- If Sports Equipment → Fixed Rate Rs. 1000 per kg

✦ **Task:**

Write a C program using nested switch-case statements to calculate total luggage fees based on type and weight

Code:

```
int main() {  
    int luggageType, subType;  
    float weight, fee = 0;  
  
    printf("Enter luggage type (1-Handbag, 2-Checked Bag, 3-Sports Equipment): ");  
    scanf("%d", &luggageType);  
  
    printf("Enter weight (kg): ");  
    scanf("%f", &weight);  
  
    switch (luggageType) {  
        case 1: // Handbag  
            printf("1-Small, 2-Large: ");  
            scanf("%d", &subType);
```

```
switch (subType) {  
    case 1:  
        fee = weight * 100;  
        break;  
    case 2:  
        fee = weight * 200;  
        break;  
    default:  
        printf("Invalid Handbag type\n");  
        return 0;  
}  
break;
```

case 2: // Checked Bag

```
printf("1-Domestic, 2-International: ");  
scanf("%d", &subType);
```

```
switch (subType) {  
    case 1:  
        fee = weight * 300;  
        break;  
    case 2:  
        fee = weight * 500;  
        break;  
    default:  
        printf("Invalid Checked Bag type\n");  
        return 0;  
}  
break;
```

```

        case 3: // Sports Equipment
            fee = weight * 1000;
            break;

        default:
            printf("Invalid luggage type\n");
            return 0;
    }
    printf("Total Luggage Fee: Rs. %.2f\n", fee);

    return 0;
}

```

Question 4: Online Bookstore Ordering System (Nested Switch-Case)

Book Categories:

- 1 → Fiction
 - 2 → Non-Fiction
 - If Fiction:
 - 1 → Novel (Rs. 600)
 - 2 → Comic (Rs. 300)
 - If Non-Fiction:
 - 1 → Biography (Rs. 800)
 - 2 → Self-Help (Rs. 500)

✦ Task:

Design a program using nested switch-case to display the selected book and calculate the total bill for given quantity.

Code:

```

int main() {
    int category, type, quantity;

```

```
int price = 0;
```

```
printf("Enter Book Category (1-Fiction, 2-Non-Fiction): ");
```

```
scanf("%d", &category);
```

```
printf("Enter Quantity: ");
```

```
scanf("%d", &quantity);
```

```
switch (category) {
```

```
    case 1: // Fiction
```

```
        printf("1-Novel, 2-Comic: ");
```

```
        scanf("%d", &type);
```

```
        switch (type) {
```

```
            case 1:
```

```
                price = 600;
```

```
                break;
```

```
            case 2:
```

```
                price = 300;
```

```
                break;
```

```
            default:
```

```
                printf("Invalid Fiction Type\n");
```

```
                return 0;
```

```
        }
```

```
        break;
```

```
    case 2: // Non-Fiction
```

```
        printf("1-Biography, 2-Self-Help: ");
```

```
        scanf("%d", &type);
```

```

switch (type) {
    case 1:
        price = 800;
        break;
    case 2:
        price = 500;
        break;
    default:
        printf("Invalid Non-Fiction Type\n");
        return 0;
}
break;

default:
    printf("Invalid Category\n");
    return 0;
}

printf("Total Bill: Rs. %d\n", price * quantity);

return 0;
}

```

Question 5: Staff Incentive Calculation (Nested Ternary Operator)

A company gives incentives based on years of employment:

- If Service > 15 years → 35% of salary
- Else if Service > 7 years → 20%
- Else → 5%

✦ **Task:**

Write a C program using a nested ternary operator to calculate and display the incentive amount.

Code:

```
int main() {  
    int years;  
    float salary, incentive;  
  
    printf("Enter years of service: ");  
    scanf("%d", &years);  
  
    printf("Enter salary: ");  
    scanf("%f", &salary);  
  
    incentive = (years >= 15) ? salary * 0.35 :  
                (years >= 7) ? salary * 0.20 :  
                salary * 0.05;  
  
    printf("Incentive Amount: Rs. %.2f\n", incentive);  
}
```

Question 6: Quadrilateral Type Identifier (Nested If-Else + Relational Operators)

User enters four side lengths of a quadrilateral. Conditions:

- If any side $\leq 0 \rightarrow$ Not a valid quadrilateral
- If all sides equal $\rightarrow$ Square
- If opposite sides equal $\rightarrow$ Rectangle
- Else $\rightarrow$ General Quadrilateral

✦ **Task:**

Design a C program using nested if-else statements and relational operators to identify the quadrilateral type.

Code:

```
int main() {  
    int a, b, c, d;  
  
    printf("Enter four sides: ");  
    scanf("%d %d %d %d", &a, &b, &c, &d);  
  
    if (a == b && b == c && c == d) {  
        printf("Square\n");  
    }  
    else if (a == c && b == d) {  
        printf("Rectangle\n");  
    }  
    else {  
        printf("General Quadrilateral\n")  
    }  
}
```

Question 7: ATM Transaction Authentication System (Logical Operators + Nested If)

System checks:

- If Card Number is valid
 - If PIN is correct
 - If Transaction Amount $\leq$ Account Balance $\rightarrow$ Transaction Successful
 - Else $\rightarrow$ Insufficient Funds
 - Else $\rightarrow$ Invalid PIN
- Else $\rightarrow$ Invalid Card

✦ Task:

Write a C program using nested if statements and logical operators (&&, ||) to validate ATM transactions.

Code:

```
int main() {
```

```
int cardValid, pinCorrect;

float balance, amount;

printf("Is card valid? (1-Yes, 0-No): ");
scanf("%d", &cardValid);

if (cardValid == 1) {
    printf("Is PIN correct? (1-Yes, 0-No): ");
    scanf("%d", &pinCorrect);

    if (pinCorrect == 1) {
        printf("Enter account balance: ");
        scanf("%f", &balance);

        printf("Enter transaction amount: ");
        scanf("%f", &amount);

        if (amount <= balance) {
            printf("Transaction Successful\n");
        } else {
            printf("Insufficient Funds\n");
        }
    } else {
        printf("Invalid PIN\n");
    }
} else {
    printf("Invalid Card\n");
}
```

Question 8: Advanced Health Calculator (Switch + Nested Conditions + Math Library)

Design a health calculator that performs:

Operations:

- 1 → BMI Calculation
- 2 → BMR Calculation
- 3 → Ideal Weight (Devine Formula)
- 4 → Heart Rate Zone (Percentage of Max HR)

Conditions:

- Weight and height must be positive
- Age must be positive for BMR and Heart Rate
- Display errors if invalid input

✦ Task:

- Use switch-case to select operation
- Use <math.h> functions like pow(), sqrt(), fabs()
- Implement nested decision structures for input validation

Code:

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    int choice, age, gender;
```

```
    float weight, height, result;
```

```
    printf("Advanced Health Calculator\n");
```

```
    printf("1. BMI Calculation\n");
```

```
    printf("2. BMR Calculation\n");
```

```
    printf("3. Ideal Weight (Devine Formula)\n");
```

```
    printf("4. Heart Rate Zone\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
switch (choice) {
```

```
    case 1:
```

```
        printf("Enter weight (kg): ");
```

```
        scanf("%f", &weight);
```

```
        printf("Enter height (meters): ");
```

```
        scanf("%f", &height);
```

```
        if (weight > 0 && height > 0) {
```

```
            result = weight / pow(height, 2);
```

```
            printf("BMI = %.2f\n", result);
```

```
        } else {
```

```
            printf("Error: Weight and height must be positive\n");
```

```
        }
```

```
        break;
```

```
    case 2:
```

```
        printf("Enter gender (1-Male, 2-Female): ");
```

```
        scanf("%d", &gender);
```

```
        printf("Enter age: ");
```

```
        scanf("%d", &age);
```

```
        printf("Enter weight (kg): ");
```

```
        scanf("%f", &weight);
```

```
        printf("Enter height (cm): ");
```

```
        scanf("%f", &height);
```

```
        if (age > 0 && weight > 0 && height > 0) {
```

```
            if (gender == 1) {
```

```

        result = 88.36 + (13.4 * weight) + (4.8 * height) - (5.7 * age);
    } else if (gender == 2) {
        result = 447.6 + (9.2 * weight) + (3.1 * height) - (4.3 * age);
    } else {
        printf("Invalid gender\n");
        break;
    }
    printf("BMR = %.2f calories/day\n", result);
} else {
    printf("Error: Invalid input values\n");
}
break;

```

case 3:

```

printf("Enter gender (1-Male, 2-Female): ");
scanf("%d", &gender);
printf("Enter height (cm): ");
scanf("%f", &height);

if (height > 0) {
    if (gender == 1) {
        result = 50 + 0.9 * (height - 152);
    } else if (gender == 2) {
        result = 45.5 + 0.9 * (height - 152);
    } else {
        printf("Invalid gender\n");
        break;
    }
    printf("Ideal Weight = %.2f kg\n", fabs(result));
}

```

```

    } else {
        printf("Error: Height must be positive\n");
    }
    break;

case 4:
    printf("Enter age: ");
    scanf("%d", &age);

    if (age > 0) {
        int maxHR = 220 - age;
        printf("Fat Burn Zone: %.0f - %.0f bpm\n",
            maxHR * 0.5, maxHR * 0.7);
        printf("Cardio Zone: %.0f - %.0f bpm\n",
            maxHR * 0.7, maxHR * 0.85);
    } else {
        printf("Error: Age must be positive\n");
    }
    break;

default:
    printf("Invalid choice\n");
}

return 0;
}

```